Do Macro-Financial Signals Improve US Equity Factor Allocation?

An exploratory study of prediction, allocation and economic states, with UK and German cross-market checks

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Abstract

This study examines whether macro-financial signals improve monthly US equity factor return forecasts and portfolio allocation, whether growth – inflation states provide incremental information, and whether similar allocation outcomes emerge in the UK and Germany. Seven US predictors are incorporated into ordinary least squares OLS, ridge and elastic net regressions estimated using expanding windows. Equally weighted combinations of their forecasts determine constrained allocations across five long-short equity factors. Supplementary cross-market analyses apply the same allocation rule to previously developed country-specific ridge forecasts. Over the 175-month US evaluation period from January 2012 to July 2026, the macro-informed strategy achieves an annualized Sharpe ratio improvement of 0.055 relative to a historical-mean benchmark constructed using the same allocation rule. However, the 95% confidence interval [−0.083, 0.145] includes zero, and the approximate minimum detectable improvement at 80% power is 0.172. For the operating profitability factor RMW, Holm-adjusted Clark-West tests indicate incremental predictive value from state indicators relative to continuous-predictor OLS models in the extended evaluation period, but not the original 88-month period. These forecasts remain weaker than the allocation mean and yield no state-allocation advantage. UK and German checks show a small, statistically unresolved UK increment and a German allocation close to its benchmark, while state portfolios have lower Sharpe estimates in all three markets. Both US strategies have negative annual arithmetic returns under a hypothetical momentum-cost charge; revised inputs, previously inspected forecasting designs, retrospective specification choices and reused validation targets make these comparisons exploratory and leave dependable investment gains unestablished.

1 Introduction

Macro-financial information can improve an allocation only if signals available before a return help make better decisions than a feasible benchmark. This study asks whether seven economic and financial variables improve next-month forecasts and allocation across US size, value, profitability, investment and momentum factors, after hypothetical reallocation charges. A second question asks whether growth–inflation states add information to the continuous predictors. The principal benchmark uses historical-mean forecasts under the same portfolio rule.

Prediction and allocation are distinct stages: forecasts can reduce errors without changing constrained weights, and portfolio gains can reflect average exposure rather than timing. This report compares both stages and transfers the frozen US mapping to UK and German forecasts, testing whether the same broad pattern appears with other inputs and factor construction.

1.1 Research questions and evaluation

The first comparison, H1, asks whether continuous signals improve predictions over a historical mean and whether the resulting Macro allocation improves portfolio performance. The second, H2, asks whether states improve continuous predictions and the associated allocation. Forecast nulls allow no positive expected adjusted-loss gain; allocation nulls set the population Sharpe difference to zero. Forecast tests assess positive adjusted loss gains for

nested OLS models; portfolio tests assess Sharpe differences, with a positive difference required for an improvement claim. Table 1 maps the questions to the tests and findings.

Table 1 Research questions and evidence

Question	Comparison	Finding
Do continuous signals improve forecasts? (H1-F)	OLS versus a mean fitted on the same past labels	No broad improvement after six-factor correction
Do continuous signals improve allocation? (H1-A)	Macro versus the identically mapped historical mean; ΔSharpe 0.055	Small positive estimate; 95% interval spans zero
Do states improve forecasts? (H2-F)	State OLS versus continuous OLS on identical labels	RMW conditional rejection in 175 months; none in 88
Do states improve allocation? (H2-A)	State portfolios versus continuous Macro	No positive gain; original fixed-state comparison deteriorates
Does the allocation pattern extend beyond the US?	Frozen allocation rule on UK and German Ridge forecasts	Small unresolved UK increment; near-zero negative German estimate; state portfolios lower

Decisions use a 5% level. US forecast tests use one-sided matched OLS Clark–West, HAC3 and separate six-factor Holm families; the 175/88 target windows have 172/85 eligible origins. Portfolio intervals use paired six-month blocks and HAC3. International allocations use country-specific Ridge forecasts and their matched historical means; their mean-test Holm family contains four country–contrast comparisons per HAC lag.

Detailed estimates, benchmark definitions and sensitivities follow in Sections 4-5; Appendix A and Table A1 define the terms and units used throughout. Appendix B records the UK and German data choices, forecast diagnostics and allocation uncertainty.

The extended window contains 175 monthly targets, January 2012 - July 2026; the original evaluation contains 88, April 2019 - July 2026. The longer window provides more observations, greater statistical resolution and wider state coverage, so it is the main descriptive window. It adds 87 previously inspected validation targets. Both windows receive adjacent headline estimates and RMW tests, because greater sample size does not create independent evidence. The comparison is pseudo-out-of-sample: forecasts are generated chronologically from earlier observations, but revised historical data, choices made after observing outcomes and validation reuse prevent an independent-holdout interpretation.

1.2 Contribution and principal findings

Macro's estimated Sharpe increment is 0.055, with a 95% interval [−0.083, 0.145]. The approximate difference detectable at two-sided 5% significance and 80% power is 0.172, about three times the observed gain. The result leaves modest benefits unresolved rather than demonstrating that the population effect is zero.

Many fitted forecasts stay close to a historical mean. RMW has conditional state-forecast evidence in the extended US window, but no matching allocation gain; Expansion is absent from the original evaluation. Absolute gains are concentrated in 2022, while the paired increment has a different annual distribution. Section 5.6 connects the UK and German forecasts to allocation performance under the frozen mapping.

2 Literature

2.1 Predictability and forecast combination

Historical averages are demanding benchmarks. Welch and Goyal (2008) document weak and unstable out-of-sample equity-premium prediction; Goyal et al. (2024) extend the evidence to additional predictors and later observations. Campbell and Thompson (2008) show that economically motivated restrictions can improve

forecasts and that modest forecast gains can matter for a particular allocation. These aggregate-market findings motivate a benchmark comparison, but next-month long - short factor forecasts require their own evidence.

Forecast combination can reduce instability across individual regressions (Rapach et al., 2010). Ridge and Elastic Net limit coefficient flexibility (Hoerl and Kennard, 1970; Zou and Hastie, 2005). When penalized fits approach an intercept-only model, their equal average also shrinks the OLS forecast toward a mean. Neely et al. (2014) find complementary macroeconomic and technical information for the aggregate equity premium, suggesting a broader information set than the seven continuous variables studied here. Their result does not establish factor-allocation gains from this smaller specification.

2.2 Factor timing and economic conditions

Fama and French (2015) define the size, value, profitability and investment spreads; Carhart (1997) incorporates momentum into a performance framework. Haddad et al. (2020) document factor predictability using valuations, principal components and expected-return restrictions. Asness et al. (2017) stress the difficulty of contrarian valuation timing, Ilmanen et al. (2021) examine signals over a much longer history, and Gupta and Kelly (2019) study factor momentum. Their different information sets and constraints caution against a general conclusion from any one timing design. Jensen et al. (2023) organize international characteristic portfolios into factor themes, enabling a broader market comparison while retaining differences from the French factor construction.

Business conditions motivate both continuous predictors and state labels. Fama and French (1989) connect expected returns with term and default spreads, while Ilmanen et al. (2014) study returns in growth and inflation environments at longer horizons. Polk et al. (2020) report gains from business-cycle factor rotation using leading indicators and global risk appetite. Their long-only factor construction and classifier differ from the US long–short factors and median-based growth-inflation states here. Bender et al. (2018) emphasize that useful predictors depend on investment horizons and that unstable relationships and hindsight complicate timing. Descriptive cyclicality therefore needs a separate test with information dated before the target return; historical revisions can alter that information (Croushore and Stark, 2001).

Dichtl et al. (2019) find predictive relations for global equity factors that are difficult to convert into net gains after turnover costs. Their portfolio-policy evidence motivates evaluating economic performance alongside forecast accuracy. Moreira and Muir (2017) show that volatility-sensitive exposure management can improve factor performance when expected returns do not increase proportionately with volatility. The present forecast-over-variance rule also uses risk information, but its normalization, shrinkage and cap form a different policy; a comparison with an identically mapped mean strategy is needed to identify the incremental role of macro forecasts.

2.3 Inference and the scope of this study

Simple allocations compete strongly with portfolios built from estimated returns (DeMiguel et al., 2009), and trading costs can alter anomaly rankings (Novy-Marx and Velikov, 2016). Sharpe inference has developed from Jobson and Korkie (1981) and Memmel (2003) to robust paired bootstrap approaches (Ledoit and Wolf, 2008). The present moving-block implementation uses that inference rationale with an influence-score HAC studentizer and block-length sensitivities.

Selection among plausible strategies creates an additional problem. White (2000) tests the best performance differential in an inspected family; Hansen (2005) offers a superior-predictive-ability refinement. Harvey et al. (2016) discuss extensive factor searches, and Bailey and López de Prado (2014) adjust Sharpe evaluation for selection and nonnormality. Here, a Reality Check covers ten documented mappings, while its bounded scope remains distinct from the wider unrecorded search.

Kelly et al. (2024) show that rich nonlinear features can improve aggregate-market timing and that forecast R^2 can be an incomplete guide to economic value. Consequently, weak performance of seven-variable linear factor forecasts does not imply that more flexible models or other information sets must fail. This report evaluates a

limited comparative question: how the chosen signals affect forecast accuracy, constrained allocation, observed state support and implementation sensitivities.

3 Data and information timing

3.1 Returns and predictors

Supplied Kenneth French monthly series contain excess-market, size, value, profitability, investment and momentum returns, denoted MKT−RF, SMB, HML, RMW, CMA and MOM (Kenneth R. French Data Library, n.d.). Values are percentage points. The primary allocation excludes MKT−RF and assigns positive weights to the five long–short spreads. Positive sleeve weights still involve short positions in the underlying stocks, and the compounded index tracks synthetic factor excess returns.

Table 2 describes the seven predictors and their proposed economic channels. Daily financial observations are averaged within the decision month. Inflation, unemployment and industrial production (IP) use the preceding reference month only when the exact observation has been released. Monthly consumer price index (CPI) inflation and IP growth require consecutive levels; missing levels remain missing. Spreads and yield changes are percentage-point quantities. The joint association of these overlapping signals is estimated separately for each factor.

Table 2 Seven predictors and their economic rationale

Predictor	Observation used	Economic channel
VIX volatility index	Decision-month daily average	Implied uncertainty and risk appetite
Baa corporate yield minus 10Y Treasury	Decision-month average spread	Credit conditions and default compensation
10Y minus 2Y yield	Decision-month average spread	Expected business conditions and term compensation
2Y yield change	Change in monthly average yield	Monetary conditions and discount-rate adjustment
CPI inflation	Monthly change of preceding released level	Price pressure and policy response
Unemployment	Preceding released monthly level	Labour-market slack and demand conditions
IP growth	Monthly change of preceding released level	Real activity and expected cash-flow conditions

Sources: supplied FRED series and the release calendar. 2Y and 10Y denote two- and ten-year Treasury maturities. Percentage-point transformations are used except the VIX index and unemployment level. Year-on-year IP and CPI define states, not replacements for continuous monthly changes.

The supplied snapshots fix the historical inputs; original factor download dates are unrecorded. The main design uses revised, current-vintage macro values with release-aware decision rules. An IP sensitivity substitutes Philadelphia Fed historical decision-month vintages while other inputs remain current vintages. Because its penalties are separately selected, that comparison changes both the IP information and the estimation settings. Appendix C also substitutes vintage IP with the main penalties held fixed to assess the input change under a common fitting policy.

3.2 Decisions, targets and missing observations

An end-of-month signal at t predicts the return ending in t+1. Training uses only return targets completed by t. Initial training comprises 261 signals, March 1990–November 2011, with targets April 1990–December 2011. Five expanding time-series folds within that sample select US penalties. Later origins refit coefficients and standardization on eligible earlier signals, retaining the selected penalties.

Late-sample exceptions are handled explicitly. The October 2025 all-items CPI level and household-survey unemployment were not published; the absent CPI level also prevents a November monthly inflation calculation (U.S. Bureau of Labor Statistics, n.d.-a). September unemployment and IP were delayed beyond relevant decisions (U.S. Bureau of Labor Statistics, n.d.-b; Board of Governors of the Federal Reserve System, n.d.). Three target months, November 2025–January 2026, therefore use historical-mean fallbacks. Matched continuous and state forecast tests have 172 eligible origins in the extended window and 85 in the original one. States have two missing labels because a year-on-year label may exist when the monthly predictor is unavailable.

Historical IP uses decision-month vintage columns from March 1990 to June 2026 (Federal Reserve Bank of Philadelphia, n.d.). Both levels in each growth calculation come from the same column, so rebasing does not become spurious growth. The November 2025 column ends in August and cannot supply the required October observation. Archived calendars and release evidence document these availability checks.

3.3 Specification choices and recorded selection

The five-factor universe, variance scaling, 50% shrinkage, 40% cap, state horizons and extended window were chosen after outcomes had been inspected. Table 3 records recoverable settings. A complete chronological search history is unavailable.

Table 3 Documented specification choices

Choice	Retained setting	Relation to observed outcomes
Earlier allocation	Raw positive forecasts across six factors	Already examined; retained as a sensitivity
Other mappings	Variance scaling, fixed market exposure, rank weights and volatility ceiling	Explored with outcomes available; list is not a complete search log
Current allocation	Five long–short factors; 36-month variances; 50% shrinkage; 40% cap	Selected after outcome inspection; frozen for the reported comparisons
State rules	Fixed and expanding year-on-year medians	Retrospective choices; neither restores Expansion in the original evaluation
Target window	175 months including the former 87-month validation period	Previously examined outcomes; not an independent holdout
Penalties	Initial training time-series cross-validation	Later validation and evaluation errors excluded from penalty selection

Exact decision dates and a complete preregistered search history are unavailable. The table records documented choices and does not claim that the listed mappings exhaust all experiments.

The selection diagnostic reconstructs ten mappings from those documented choices, fixing their definitions before the additional test. Each applies the same continuous forecasts and compares them with historical means under its own mapping. Appendix D reports all ten together. A secondary family adds both state models, giving thirty candidates. The correction covers these recorded families; earlier unrecorded choices and the reuse of validation data remain outside it.

3.4 UK and German data and transfer design

The supplementary check uses five JKP value-weighted USD excess-return themes of size, value, investment, profitability and momentum (Jensen et al., 2023; JKP Global Factor Data, n.d.), alongside CPI and industrial production from ONS for the UK and Destatis for Germany (Office for National Statistics, n.d.-a, n.d.-b; Federal Statistical Office of Germany, n.d.-a, n.d.-b), and the ECB country-level financial-stress index CLIFS (European Central Bank, n.d.). Theme names align with broad US factor styles, but JKP aggregates signed characteristic portfolios and does not reproduce the French factors. Table B2 records the retained sources and documented candidate choices.

Existing international models use year-on-year IP growth and CLIFS at two-month reference lags, and CPI inflation at one month. These availability proxies differ from the US monthly signals and exact release checks; a complete historical release audit is unavailable. Four initial time-series folds select Ridge penalties separately by country, theme and continuous or fixed-state model. Later refits retain those penalties. UK training has 309 signals and evaluation has 133 targets, December 2014–December 2025; Germany has 283 and 122, November 2015-December 2025.

Allocation retains those forecasts and transfers the US mapping: 36-month variance scaling, 50% equal-weight shrinkage, a 40% cap and a 10 bp target-weight turnover charge. Local historical means use the same completed training labels. The forecasting designs were previously inspected; the retrospective transfer introduces no further penalty or mapping selection.

4 Methods

4.1 Forecasts and benchmark means

OLS, Ridge and Elastic Net (EN), each with an unpenalized intercept, predict the next completed return from the seven signals. Scaling uses eligible training observations only. Their equal average forms the conditional forecast:

$$\hat{\mu}_{j,t+1|t} = \frac{\hat{\mu}_{j,t+1|t,\text{OLS}} + \hat{\mu}_{j,t+1|t,\text{Ridge}} + \hat{\mu}_{j,t+1|t,\text{EN}}}{3} \quad (1)$$

Initial cross-validation minimizes mean squared error (MSE). The Ridge grid has 41 penalties from 10^{-4} to 10^{6}; EN has 26 from 10^{-4} to 10 and mixing ratios 0.1, 0.5 and 0.9. US state models reuse the continuous-model penalties; the historical-IP sensitivity is separately tuned. Later refits update coefficients and scalers without retuning on evaluation returns.

Four of six Ridge fits select 10^{6}, with initial slope effective degrees of freedom near 0.0018; five EN fits have zero slopes. Among the five allocated factors, three Ridge fits reach the boundary and four EN fits have zero slopes. For these boundary factors, the average is close to the training mean plus one-third of the OLS departure. SMB and MOM retain more flexibility. Effective degrees of freedom measure fitted-response flexibility; an EN active-coefficient count is a simpler complexity proxy. Appendix D reports both, while the calculations preserve all three actual fitted predictions.

Two historical means serve distinct purposes. The allocation benchmark averages all completed factor returns from April 1990 through the signal month. For nested forecast tests, continuous OLS instead uses an intercept-only benchmark fitted to exactly the same eligible labels; state OLS uses reduced continuous OLS on exactly the augmented model's labels. Table 4 compares forecasts descriptively with the allocation mean. Matched tests compare models trained on common observations, avoiding gains caused by changing the training sample.

4.2 Growth–inflation states

States use preceding-reference-month year-on-year IP growth and CPI inflation. Values strictly above their initial medians are high; equality is low. High growth/low inflation is Expansion, high/high Overheating, low/low Slowdown and low/high Stagflation. Initial current-vintage thresholds are 2.84% and 2.79%. Three indicators augment the continuous models, with Expansion as reference category.

The fixed rule retains these medians. The expanding rule starts from the same 261 labels, computes later thresholds from earlier signal values and excludes the current observation. Previously assigned labels stay fixed during refitting. Thresholds update independently, but a current state requires both inputs. Long initial histories can make an expanding median slow to adapt.

4.3 Allocation and hypothetical costs

Forecasts are divided by trailing 36-month variances of completed factor returns. Negative scores become zero and positive scores are normalized; all nonpositive scores trigger equal weights. The vector is shrunk halfway toward five-factor equal weights and capped at 40%:

$$q_{j,t} = \max\left(0, \frac{\hat{\mu}_{j,t+1|t}}{\hat{\sigma}_{j,t,36}^{\,2}}\right), \qquad p_{j,t} = \frac{q_{j,t}}{\sum_k q_{k,t}} \ \text{ if } \sum_k q_{k,t} > 0 \quad (2)$$

$$w_t = \mathrm{C}_{0.40}\left(0.5p_t + \frac{0.5}{5}1\right) \quad (3)$$

The capping operator C in equation (3) fixes overweight factors at 40% and redistributes excess proportionally until weights are feasible. Weights sum to one and range from 10% to 40%. Macro and historical means share this mapping. Equal weights and inverse individual volatility are simple benchmarks; Appendix D includes other recorded mappings.

The main charge is 10 basis points (bp) times one-way target-weight turnover, half the sum of absolute weight changes. One bp is 0.01 percentage point. Equation (4) uses decimal returns and charge c; each evaluation window begins with zero initial turnover. Scenarios at 0, 25 and 50 bp retain the weights. Target-weight turnover omits intervening holdings drift and differs from actual rebalancing trades.

$$r_{\mathrm{net},t+1} = \sum w_{j,t} r_{j,t+1} - \frac{c}{10{,}000} \times \frac{1}{2}\sum \left| w_{j,t} - w_{j,t-1} \right| \quad (4)$$

A separate scenario charges 65 bp monthly per unit of each strategy's MOM weight. Novy-Marx and Velikov (2016) motivate heterogeneous internal factor costs, but these 65 bp and 10 bp assumptions are not calibrated to the French portfolios. Reallocation charges omit within-factor stock trades, borrowing and financing. The MOM scenario illustrates one further exposure-dependent burden.

4.4 Forecast and portfolio inference

Forecast R^2 is one minus model MSE divided by benchmark MSE. For nested US OLS, Clark–West adds the squared forecast difference to benchmark-minus-model squared loss, adjusting for estimation-induced bias, and tests a positive expected gain (Clark and West, 2007). Ordinary loss differences use a two-sided heteroskedasticity and autocorrelation consistent (HAC) reference (Diebold and Mariano, 1995; Newey and West, 1987); recursive nesting complicates that reference (Clark and McCracken, 2001). Holm (1979) adjusts the six US factors within each model, window, test and HAC lag. The primary lag is three months, with six and twelve as sensitivities. Applying Clark–West to shrinkage or averaged forecasts is an approximate extension of its nested OLS setting.

Sharpe is monthly mean divided by sample standard deviation, annualized by $\sqrt{12}$. Paired mean-return tests use Bartlett HAC covariance and Holm correction for the three portfolio comparisons in each window. Sharpe intervals use 3,000 paired moving-block draws, a primary six-month block and a three-lag HAC studentizer. Delta-method influence scores estimate the paired Sharpe standard error; each bootstrap draw recomputes it, and percentile-t quantiles form the interval. Three- and twelve-month blocks assess dependence sensitivity (Künsch, 1989; Ledoit and Wolf, 2008). Predictions and decisions remain fixed during resampling. The cross-market check uses the same paired procedure; international mean-return tests form a separate four-comparison Holm family within each HAC lag (two countries, two contrasts).

$$\mathrm{SR} = \sqrt{12} \times \frac{\overline{r}}{s_r} \quad (5)$$

$$\mathrm{MDE} \approx (z_{0.975} + z_{0.80})\mathrm{SE}_{\Delta SR} \approx 2.8016\mathrm{SE}_{\Delta SR} \quad (6)$$

In equations (5)–(6), the overbar is a monthly mean, s a sample standard deviation, z a standard-normal quantile and SE the paired HAC standard error. The minimum detectable effect (MDE) is a local normal approximation for two-sided 5% significance at 80% power. Its scale concerns the fitted strategy paths. Constraints can reduce both observed differences and their standard errors, so a smaller MDE alone does not establish better information.

4.5 Selection, attribution and sensitivity

The White-style Reality Check uses the ten mappings' net Macro-minus-mean returns (White, 2000). Its null is that all expected differentials are nonpositive. The unstudentized statistic is the maximum of zero and √N times the largest mean. Candidates are centered at their own means to impose a boundary null; common non-circular overlapping blocks preserve pairing across candidates. The primary calculation uses six-month blocks and 9,999 draws with a plus-one tail probability. Appendix D reports other block lengths and the thirty-candidate family. This adjusts recorded mapping selection conditional on fitted forecasts and evaluation windows.

Exposure accounting separates average-weight returns from deviations around those weights. Full-window average exposures use later decisions, so this is retrospective attribution. A spanning regression measures residual average Macro return relative to the historical-mean strategy. Event and single-month exclusions retain original weights and charges; resampling respects retained contiguous segments. Cost crossings similarly retain paths and identify the hypothetical charge at which the arithmetic mean differential reaches zero.

5 Results

5.1 Forecast accuracy and initial-sample associations

Continuous forecasts have small descriptive gains without a corrected rejection against matched historical means. Table 4 includes all 175 targets and the three fallbacks; the equal average has positive R^2 for three factors. No six-factor matched OLS Clark–West family rejects at HAC lags three, six or twelve. The smallest primary-lag Holm p-value is 0.433 in the extended window and 0.404 in the original window.

Table 4 Forecast accuracy against the allocation historical mean

Factor	OLS R^2 %	Ridge R^2 %	Elastic Net R^2 %	Average R^2 %
MKT−RF	−4.63	+<0.01	+<0.01	0.16
SMB	−6.60	1.06	0.17	−0.94
HML	0.38	−0.01	−0.02	1.46
RMW	−6.25	0.01	0.01	−1.55
CMA	−0.65	−<0.01	−0.01	0.32
MOM	−6.42	0.59	+<0.01	−0.18

175 targets including three mean fallbacks. Every column uses the same allocation historical mean; positive R^2 means lower MSE. Matched-label OLS tests appear separately.

Figure 1 traces accumulated benchmark-minus-model squared forecast errors: positive values indicate smaller model errors. Continuous OLS uses the matched intercept on 172 eligible origins; the average forecast uses the allocation mean on all 175 targets, including fallbacks. Skipped OLS dates hold the cumulative total unchanged. These raw loss paths show when gains and losses accumulate; they are distinct from the adjusted Clark–West statistic.

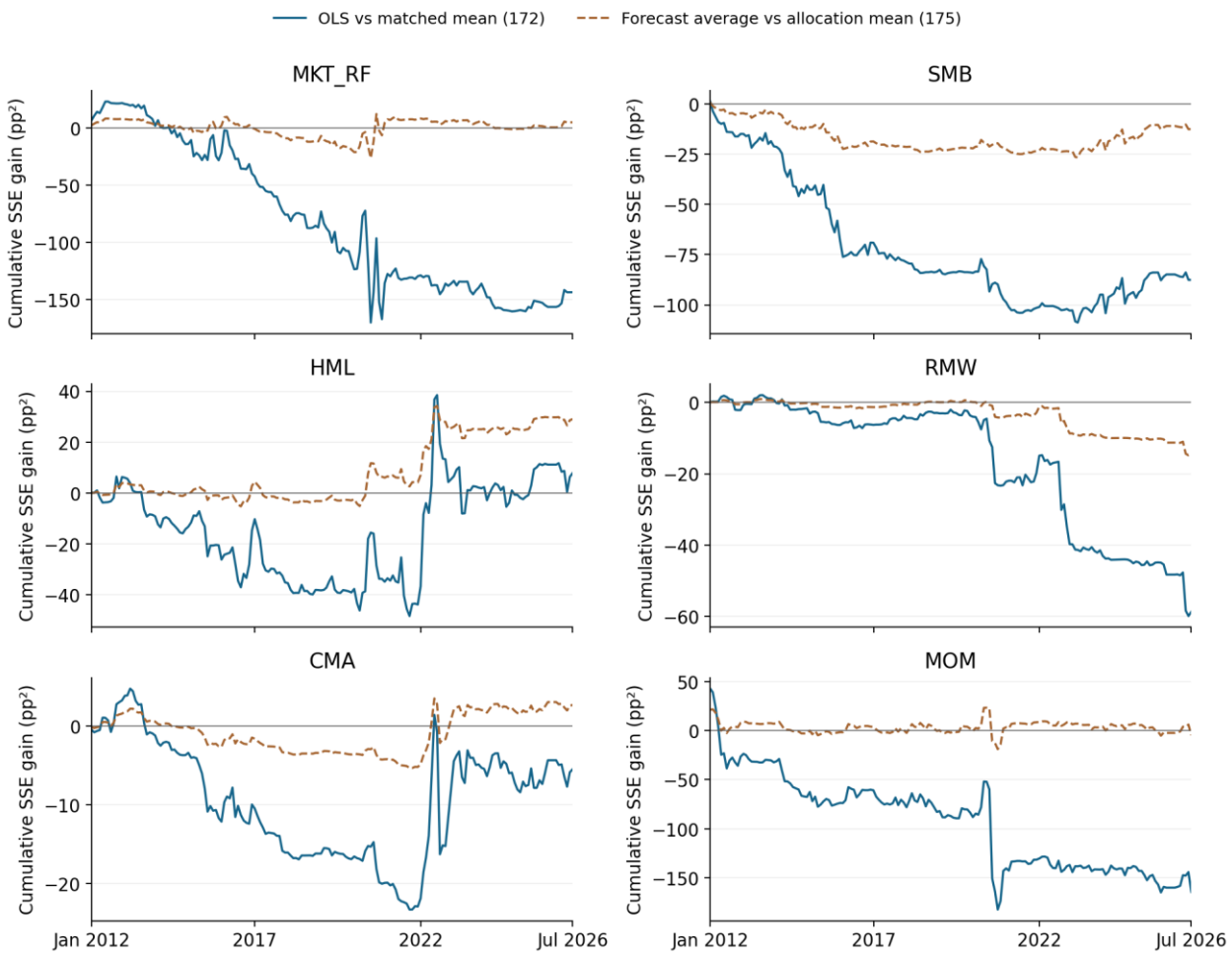


Figure 1 Cumulative raw squared-error gains for six factor forecasts. Solid lines compare continuous OLS with its matched-label mean across 172 eligible targets; dashed lines compare the actual three-model average with the allocation mean across all 175 targets, including three fallbacks. Positive values indicate lower squared error. The different benchmarks and target sets are identified explicitly; these lines contain no Clark–West adjustment and are not significance tests.

The average's HML R^2 exceeds each component, consistent with penalized fits tempering an unstable OLS forecast. Appendix D measures initial complexity. Appendix E links Table 2's proposed channels to training associations: four of 42 OLS slopes have raw HAC $p<0.05$, but only SMB–credit spread survives Holm correction. This limited evidence is consistent with strong shrinkage, although slope tests do not explain the cross-validated penalty choice or establish later predictive skill.

RMW rejects under specific adjusted state-forecast comparisons. Fixed-state OLS has Clark–West Holm $p=0.0066$ over 175 months and expanding-state OLS $p=0.0150$. The original 88-month window does not reject, with $p=0.0618$ and $p=0.244$, respectively. Fixed-state ordinary-loss Holm $p=0.1852$ in the extended window also does not reject. Adjusted and ordinary loss tests use different statistics and directional alternatives.

Across 172 eligible targets, reduced continuous RMW OLS has $R^2=-6.27\%$ against the allocation mean; fixed and expanding state OLS have −2.91% and −3.17%. Thus the state result improves a weaker OLS model while remaining less accurate than the fuller mean benchmark. RMW measures operating profitability rather than the broader quality concept. In the original window's 86 labeled target months under the fixed rule, subsequent RMW averages 2.041% in Overheating (11 observations), −0.087% in Slowdown (39) and 0.116% in Stagflation (36), with no Expansion observation. Appendix E reports conditional means and sample sizes under both rules and across windows. The largest observed mean occurs in Overheating; these associations do not establish defensiveness, NBER recession exposure or a causal explanation of the forecast improvement.

5.2 Allocation performance and statistical resolution

Macro has the largest displayed point Sharpe. Equal weighting gives annual arithmetic mean 0.65% and Sharpe 0.145. Macro has Sharpe 0.258 versus 0.203 for historical means, annual mean 1.18% and synthetic compounded rate 1.08% (Table 5). Figure 2 displays cumulative paths and the smaller additive paired increment. Monthly target-weight turnover is 5.5% for Macro and 1.1% for historical means.

Table 5 Primary allocation over 175 target months

Strategy	Mean %	Volatility %	Sharpe	CAGR %	Max DD %
Equal weight	0.65	4.50	0.145	0.55	−14.88
Inverse volatility	0.74	4.45	0.165	0.64	−14.49
Historical mean	0.88	4.33	0.203	0.79	−12.98
Macro	1.18	4.57	0.258	1.08	−12.38
Fixed state	1.09	4.63	0.234	0.98	−11.80
Expanding state	1.07	4.67	0.229	0.96	−12.45

January 2012–July 2026; five long–short factors; current-vintage inputs; 10-basis-point allocation charge. Mean and volatility are annualized. CAGR and drawdown use synthetic compounded excess returns and omit funding and factor-internal trading.

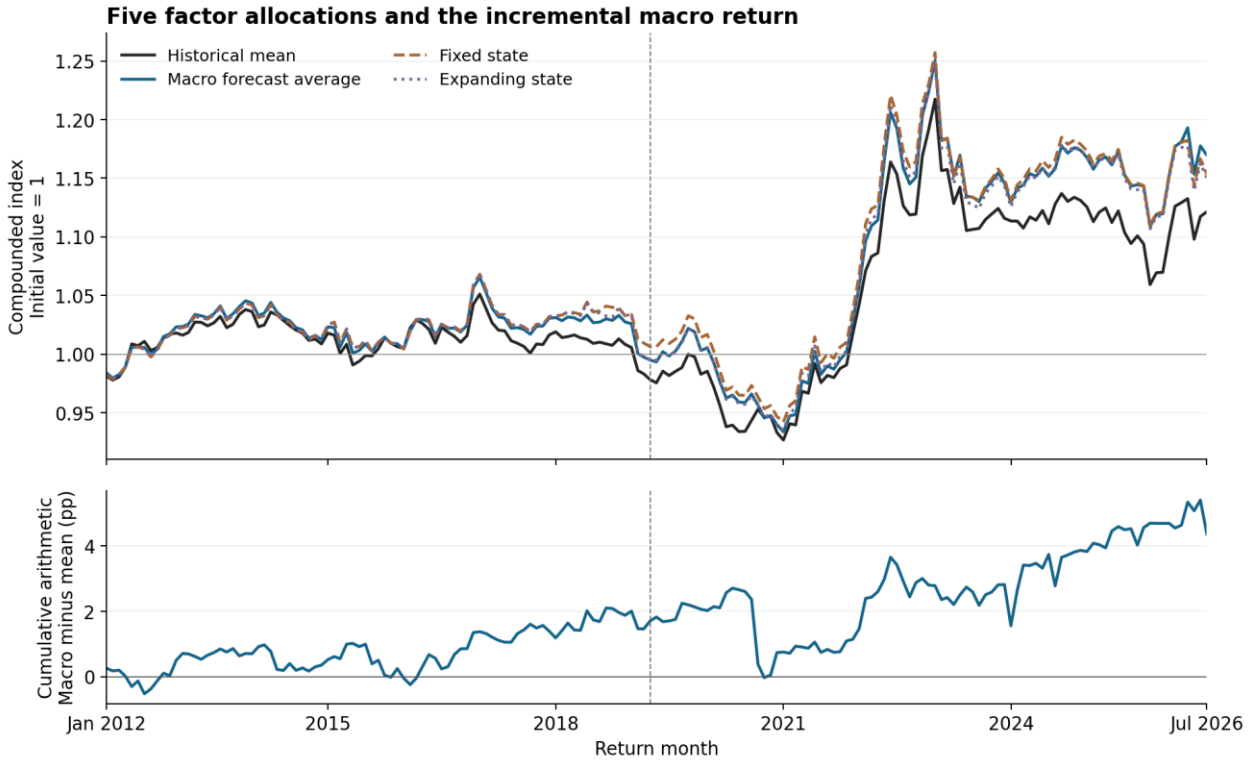


Figure 2 Primary synthetic excess-return paths and cumulative Macro-minus-historical-mean arithmetic difference. All strategies use 10 bp allocation charges. The dashed boundary begins the original April 2019 target window. The lower panel sums paired monthly arithmetic differences and is distinct from compounded wealth differences.

The annual arithmetic increment is 0.300 percentage points (pp), with HAC p=0.292 and Holm p=0.877 for the three portfolio comparisons. The Sharpe increment is 0.055, with 95% interval [−0.083, 0.145]. Table 6 also shows the original 88-month estimate and interval, which spans zero. The primary allocation comparison does not reject.

Table 6 Paired portfolio differences and uncertainty

Comparison	N	Δ Sharpe	95% interval	MDE	Mean gain pp	Holm p
Macro minus historical mean	175	0.055	[−0.083, 0.145]	0.172	0.300	0.877
Fixed state minus Macro	175	−0.023	[−0.104, 0.027]	0.086	−0.093	0.877
Expanding state minus Macro	175	−0.029	[−0.101, 0.014]	0.080	−0.109	0.877
Macro minus historical mean	88	0.041	[−0.155, 0.158]	0.236	0.364	0.653
Fixed state minus Macro	88	−0.064	[−0.179, −0.006]	0.105	−0.341	0.459
Expanding state minus Macro	88	−0.043	[−0.158, 0.016]	0.104	−0.226	0.653

Sharpe intervals: 3,000 paired moving-block percentile-t draws, six-month blocks, HAC3 studentization, net 10-basis-point allocation charges. MDE: approximate annual Sharpe difference detectable at two-sided 5% and 80% power. Mean gains are annual percentage points; Holm adjusts three mean comparisons within each window. No correction covers the wider specification search.

The MDE of 0.172 is about three times the observed Sharpe increment. Figure 3 shows how the mapping confines strategies to related exposures. It limits extreme estimated-return bets, while affecting both measured gain and uncertainty. The sample consequently has limited resolution for an effect this small.

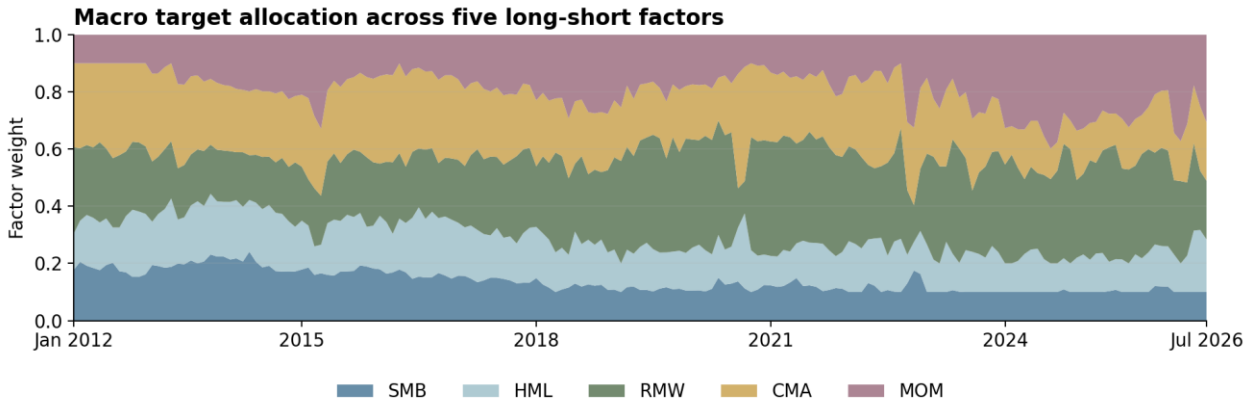


Figure 3 Macro weights in the five long–short factors. Dates mark return targets; weights use information at the preceding decision month. The market-factor weight remains zero.

5.3 State coverage and allocation outcomes

Both state allocations have lower Sharpe point estimates than continuous Macro over 175 targets, with intervals spanning zero for all reported block lengths. In the original window, the fixed-state six-month interval is below zero; the expanding-state interval spans zero at six months but is [−0.154, −0.006] at twelve. State allocation offers no positive improvement, and some dependence settings indicate deterioration.

Table 7 and Figure 4 show the original window's coverage failure. Mean IP year-on-year growth falls from 2.05% in initial training to −0.16% across 86 available evaluation signals. Only eleven later signals exceed the fixed 2.84% threshold, all with high inflation, leaving zero Expansion observations under either rule. The expanding growth median falls to 2.25% at the final signal, yet its long initial history does not restore that state.

Table 7 State coverage on signal dates

Rule and window	Expansion	Overheating	Slowdown	Stagflation	Missing	Total
Fixed / Training	75	55	56	75	0	261
Fixed / Validation	21	5	59	2	0	87
Fixed / Evaluation	0	11	39	36	2	88
Expanding / Training	75	55	56	75	0	261
Expanding / Validation	22	7	56	2	0	87
Expanding / Evaluation	0	11	31	44	2	88

Training: March 1990–November 2011. Former validation: December 2011–February 2019. Original evaluation: March 2019–June 2026. Each return target ends one month later. Counts use unique calendar signals, not factor rows.

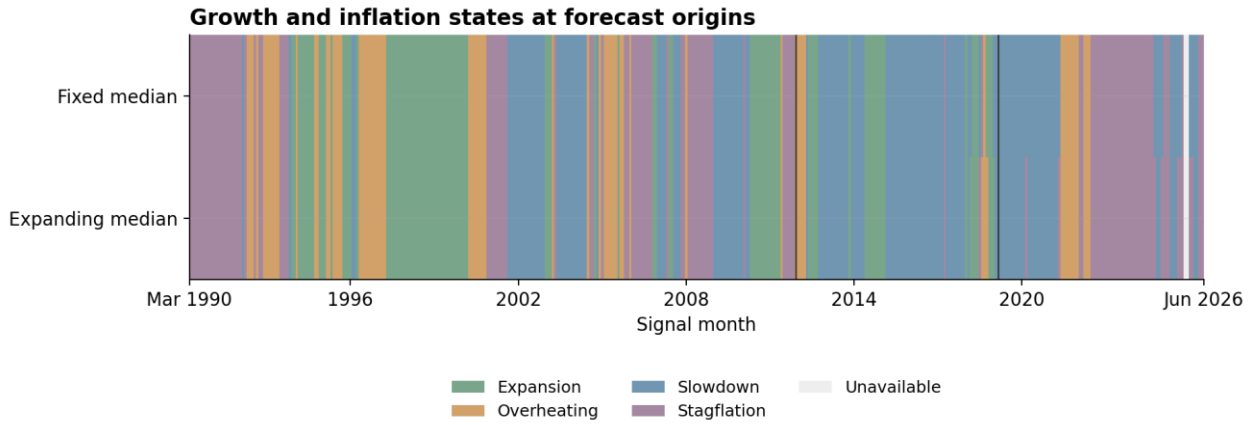


Figure 4 Growth–inflation states on signal dates under fixed and expanding medians. Boundaries separate initial training, former validation and original evaluation. Missing observations remain visible.

The extended window has 21 fixed-rule and 22 expanding-rule Expansion observations, all from reused validation months. Original-window absent-state means remain undefined in Appendix E. The observed state models can still be compared, but the original window cannot evaluate all four states on equal empirical support.

5.4 Costs, IP vintages and mapping dependence

Table 8 reports cost sensitivities; Appendix C separates IP inputs and estimation settings. With no reallocation charge, the primary mean-gain HAC3 p-value is 0.2122, so transaction costs are not the sole reason that improvement remains unresolved. On frozen paths, the arithmetic increment reaches zero at approximately 66.6 bp per unit of one-way sleeve turnover when only the allocation charge varies, or at 148.8 bp monthly per unit of MOM exposure when the sleeve charge stays at 10 bp. These are mean-return crossings, not Sharpe crossings, significance thresholds or calibrated trading costs.

Under the separate 65 bp monthly MOM charge, annual arithmetic returns are −0.36% for Macro and −0.53% for historical means. A positive relative difference can therefore coexist with negative absolute performance. Costs matter for interpretation even though the uncharged primary path does not establish an advantage.

Table 8 Cost sensitivity over 175 months

Scenario	Historical mean Sharpe	Macro Sharpe	Δ Sharpe	Mean gain pp
0 bp allocation charge	0.206	0.272	0.066	0.353
25 bp allocation charge	0.199	0.236	0.037	0.220
50 bp allocation charge	0.191	0.200	0.008	0.088
10 bp plus MOM 65 bp	−0.122	−0.079	0.044	0.169

Allocation charges are basis points per half absolute target-weight change. The momentum scenario adds a hypothetical 65-basis-point monthly charge per unit MOM weight to each strategy's own weights. Mean gain is Macro minus historical mean in annual percentage points. IP input and penalty comparisons appear in Appendix C.

The separately retuned historical-IP design raises the annual mean increment to 0.449 pp and the Sharpe increment to 0.089, with interval [−0.015, 0.178]. Mean-return HAC p-values are 0.066, 0.040 and 0.016 at lags three, six and twelve; corresponding Holm values are 0.198, 0.121 and 0.047 within each archived three-portfolio-comparison family. The primary lag does not reject, whereas the twelve-lag sensitivity does; the six-block Sharpe interval still includes zero. Conclusions therefore depend on the metric and dependence setting.

IP substitution also changes the selected HML Ridge penalty from 10^6 to 10^4. Holding the main penalties fixed gives an annual mean increment of 0.445 pp, close to the separately retuned 0.449 pp. Retuning contributes 0.004 pp of the combined 0.150 pp increase over the current-vintage design, so the penalty change is a small part of this observed sensitivity. Appendix C reports the controlled comparison and inference; it substitutes IP information under common fitting and portfolio settings while other macro series and factor returns remain current vintages.

The forecast-to-weight mapping also matters. Unscaled, uncapped five-factor positive forecasts have ΔSharpe=0.221 and monthly turnover 13.8%, compared with 0.055 and 5.5% under the main mapping. Variance scaling, shrinkage and the cap change together, producing a smaller point gain and lower turnover; the contrast does not identify each restriction's separate contribution. Appendix D retains all ten mappings. Their primary Reality Check p-value is 0.090. The family null is not rejected at 5%. The thirty-candidate and block-length sensitivities assess the same bounded selection problem.

Retrospective accounting attributes 0.339 annual pp of the gross Macro-minus-mean difference to changing weights and 0.014 pp to average exposure. Charges leave the 0.300 pp net increment. The spanning intercept is 0.282 annual pp, with HAC p=0.307, offering no reliable residual mean under this comparison.

5.5 Absolute return concentration and incremental gains

Absolute portfolio returns are concentrated in 2022: Macro compounds by 18.22% that year and −1.06% over all other observed months combined; historical means produce 16.78% and −3.97%, respectively. The allocation increment has a different distribution. Of the 4.37 pp summed monthly Macro-minus-mean difference, 2022 contributes 1.32 pp, or 30.1%, while 2024 contributes 2.28 pp. Excluding 2022 leaves a 0.225 pp annual arithmetic increment. Figure 5 separates annual compounded portfolio returns from additive paired differences; 2026 covers January–July. Separate compounded wealth differences are not additive yearly contributions.

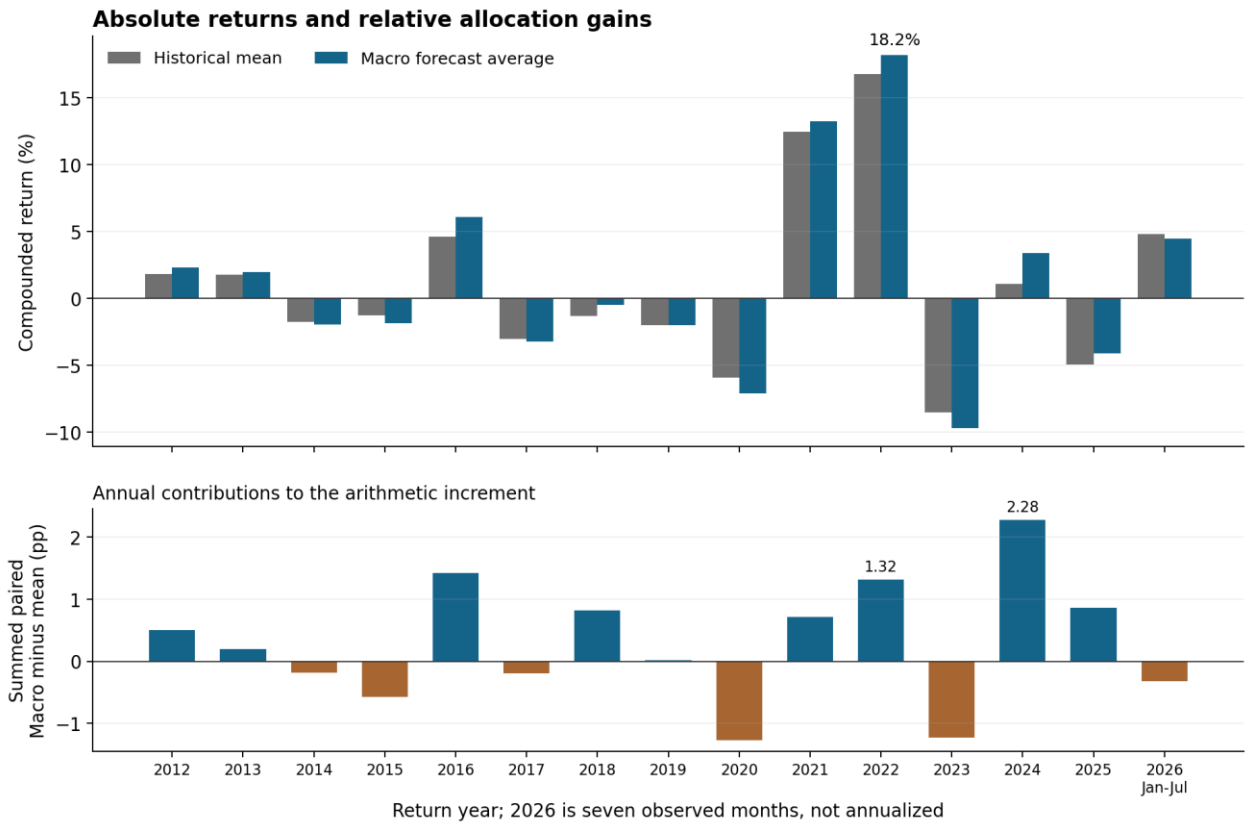


Figure 5 Annual absolute and incremental returns under the primary allocation and 10 bp charges. The upper panel separately compounds Macro and historical-mean returns; the lower panel sums paired monthly Macro-minus-mean differences in pp. The lower bars add to the full arithmetic increment. The 2026 bars cover January–July only. Absolute return concentration in 2022 differs from the annual pattern of relative gains.

Event results depend on the mapping. From March–December 2020, summed net Macro-minus-mean differences are −1.36 pp under the main five-factor rule and 15.97 pp under uncapped six-factor forecasts. Excluding these outcomes raises the main annual mean increment to 0.417 pp, while the raw six-factor increment becomes approximately zero (Appendix D). These exclusions retain the original decisions and charges.

Removing August 2020 alone raises the main Sharpe increment from 0.055 to 0.086. The same month is removed from both frozen paths; its nearly −1.99 pp paired increment explains its influence. July 2026 contains RMW +11.05% and MOM −12.25%, illustrating further variation in the factor inputs. Such outcome deletions describe sensitivity and do not identify an improved strategy or justify excluding unfavorable months from the main result.

5.6 UK and German cross-market allocation check

Table 9 answers the allocation question on the inherited country forecasts, using the frozen US mapping without additional tuning. Macro is compared with a local historical-mean strategy under that same mapping; the fixed-state strategy is compared with continuous Macro. Estimates are paired within each market, not pooled across differing factor definitions, currencies and evaluation windows.

Table 9 UK and German allocations under the frozen rule

Market / N	Equal weight SR	Mean SR	Macro SR	Δ SR	95% interval	Mean gain pp	HAC3 p
UK / 133	0.936	0.730	0.791	0.060	[−0.070, 0.264]	0.098	0.475
DE / 122	0.588	0.598	0.598	−0.00029	[−0.00404, 0.00255]	−0.0022	0.676

UK: December 2014–December 2025; Germany: November 2015–December 2025. Each market uses five JKP themes, 36 prior calendar months of variance, 50% equal-weight shrinkage, a 40% cap and a hypothetical 10 bp sleeve charge. Differences compare Macro with its identically mapped, matched-label historical mean and are computed before rounding strategy levels; mean gain is annual pp. Sharpe intervals use 3,000 paired six-month moving-block percentile-t draws, HAC3 studentization, and condition on fitted paths. Mean p-values are two-sided; four-comparison international Holm p=0.951 for both Macro–mean contrasts. Equal weight is a separate benchmark. No within-theme cost calibration is imposed.

Both Macro-minus-mean Sharpe intervals in Table 9 include zero, and neither primary-lag mean-return comparison rejects. The UK increment, 0.060, resembles the small US estimate; Germany's −0.00029 is close to zero. The UK MDE is 0.182, around three times its point increment. UK equal weighting has Sharpe 0.936, exceeding Macro's 0.791. Any benefit relative to historical means remains uncertain and does not extend to superiority over simple equal weighting.

Mean half-absolute Macro-minus-mean weight distances are 3.82% in the UK and 0.13% in Germany, compared with 8.00% in the US. Eleven of twenty inherited international Ridge fits select the largest penalty and depart from their matched training means by at most 0.1 bp in root-mean-square terms. German shrinkage therefore gives little forecast variation for the mapping to convert into differentiated weights. Its small MDE of 0.00451 concerns these nearly coincident paths and does not show that macro information is easier to detect.

Fixed-state portfolios have Sharpe differences of −0.097 in the UK and −0.019 in Germany relative to continuous Macro. The UK six-month-block interval [−0.528, −0.002] is below zero, but the twelve-month interval [−0.5561, 0.0005] crosses it; Germany's primary interval also includes zero. These are marginal fixed-path intervals, without a four-comparison Sharpe correction. State models use separate initial-training penalties and Slowdown as the omitted category, whereas US state models reuse continuous penalties and omit Expansion. Table B3 reports dependence sensitivity and all four states' support. The cross-market comparisons offer no positive state-allocation gain, while country differences do not isolate market effects or prove that useful macro information is absent.

6 Discussion and conclusion

6.1 What the comparisons establish

The chosen continuous signals produce a small positive allocation estimate with uncertainty wider than the gain. Corrected matched forecast tests do not establish improvement over historical means, and the primary Sharpe interval spans zero in both windows. An MDE near 0.172 explains why a Sharpe increment near 0.055 is difficult to distinguish from variation in these fitted paths. This is limited resolution rather than proof of a zero effect.

RMW provides narrower state evidence: adjusted OLS forecast tests reject in the extended window but not the original one, and both augmented forecasts remain weaker than the allocation historical mean. State portfolios do not improve on continuous Macro. Missing Expansion in the original window further limits the four-state interpretation. The result is a conditional forecast improvement whose portfolio value remains unestablished.

Training shrinkage and portfolio constraints leave related exposures and modest strategy differences. Initial slope tests support few associations, but do not explain why particular penalties were selected. Both portfolios' absolute gains are concentrated in 2022; their relative difference has a broader annual pattern. Separately retuned IP and alternative mappings give larger estimates, with conclusions sensitive to dependence settings and portfolio design.

These qualifications describe the evidence without generalizing to all macro information or all factor-timing methods.

The international transfer finds a small uncertain UK gain and nearly coincident German Macro and mean weights. Lower state-portfolio Sharpe estimates across three markets repeat the descriptive pattern, under differing forecasting designs. This highlights how shrinkage can prevent forecast signals from materially changing allocation.

6.2 Limitations and a subsequent test

Retrospective selection is the central limitation. Predictors, mapping choices, state definitions and the extended window were inspected with outcomes available, and the full search history is unavailable. The Reality Check covers the recorded mappings; Holm corrections and bootstrap intervals condition on their specified families and fixed generated paths. They do not include unrecorded selection or model-refitting uncertainty. A subsequent evaluation should freeze one specification before collecting new target returns.

Information and implementation limits also remain. Historical IP is reconstructed, but other inputs and factor returns use supplied current vintages, original factor retrieval dates are unknown, and some historical latency is unrecorded. Synthetic long–short returns omit actual holdings, collateral, borrowing and financing. Target-weight turnover omits drift and within-factor trades. The cost assumptions expose sensitivity but cannot establish investability.

International comparability is limited by three-predictor Ridge forecasts, different theme construction and lag proxies, shorter samples and prior inspection of both countries. JKP returns are USD excess returns; no currency-hedged counterpart is supplied. Different theme and portfolio weights can retain different currency-related exposures, so a paired comparison cannot be assumed to remove exchange-rate effects. The 10 bp sleeve charge omits internal theme trades and is hypothetical in every market.

A future study could use a fully real-time predictor panel and explicit holdings, then compare a small prespecified economic alternative on new data. A growth-change state or rolling threshold might improve state support, while macro and technical combinations or nonlinear models might use information differently. Adding such variants to the inspected sample would answer a further exploratory question. For this signal set and allocation rule, the present evidence supports a small uncertain gain, conditional RMW prediction evidence and no dependable state-allocation advantage.

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Appendices

Appendix A Terms and conventions

Table A1 Terms used in this report

Term	Meaning in the experiment
Signal and target	Month-end information and the return ending in the following month
Historical mean	Average of completed past factor returns; matched-label means are identified separately
Factor weight	Share of a synthetic long–short factor return; not a long-only stock holding
Fallback	Historical-mean forecast used when a required current predictor is unavailable
HAC	Covariance estimate allowing heteroskedasticity and dependence across specified calendar lags
HAC3 / HAC6 / HAC12	HAC covariance uses 3, 6 or 12 calendar-month lags; these are not bootstrap block lengths
Clark–West test	One-sided adjusted squared-loss comparison for a nested forecast model; penalized and ensemble applications are exploratory approximations
Sleeve	An allocation to one synthetic factor or theme return; sleeve turnover excludes trading within its long and short stock portfolios
Holm adjustment	Sequential multiplicity correction within an explicitly stated comparison family
Studentized interval	Bootstrap interval formed from an estimate divided by a recomputed standard error
Percentile-t	Quantiles of bootstrap estimate departures divided by their recomputed standard errors
MDE	Local approximate difference detectable at a specified significance level and power
Reality Check	A joint bootstrap comparison of the largest matched mean-return gain in a stated candidate family
Spanning regression	Macro strategy returns regressed on historical-mean strategy returns and an intercept

Returns and differences are labelled explicitly: % for return levels, pp for percentage-point differences, and bp for basis points. One percentage point equals 100 basis points. Sharpe ratios and their differences are unitless annualized summaries.

Appendix B UK and German cross-market checks

Table B1 retains the country-level prediction evidence; Table B2 documents source choices and definitions; Table B3 connects allocation uncertainty and state support to the main cross-market comparison. The five JKP themes are a thematic comparison with the US spreads, rather than identical portfolios. Original USD series are retained and no exchange-rate hedge is inferred.

Table B1 UK and German Ridge forecasts

Country and theme	Macro R^2 %	State R^2 %	Macro RMS bp	State RMS bp
UK size	1.37	+<0.01	13.43	0.02
UK value	−<0.01	−0.16	<0.01	1.11
UK investment	−0.65	−0.72	6.97	7.37
UK profitability	−<0.01	−<0.01	<0.01	0.01
UK momentum	−0.09	−0.70	1.08	9.30

Country and theme	Macro R^2 %	State R^2 %	Macro RMS bp	State RMS bp
DE size	+<0.01	0.24	<0.01	1.18
DE value	+<0.01	−<0.01	0.01	0.02
DE investment	+<0.01	−<0.01	<0.01	0.01
DE profitability	−<0.01	−<0.01	<0.01	<0.01
DE momentum	−0.04	−0.84	1.43	12.65

UK 133 and Germany 122 targets; IP reference lag two, CLIFS lag two, CPI lag one. R^2 compares local historical means. RMS is the prediction's departure from its matched training mean, not its return forecast error. Signed R^2 values smaller than 0.01% are shown as +<0.01 or −<0.01.

Table B1 compares Ridge forecasts with local historical means on common target dates. RMS departures measure prediction variation around each fit's training mean, not forecast errors. Continuous and state models each retain their own initial-training penalties. The fixed-state design uses the initial medians of lagged year-on-year growth and inflation, no state smoothing, and Slowdown as the reference category; thresholds remain fixed at later refits.

Table B2 International data sources and inclusion decisions

Input	Source	Use and documented constraint
UK CPI and IP	ONS D7BT / K222	All-items CPI and seasonally adjusted production levels; year-on-year changes; CPI reference lag one and IP lag two.
German CPI and IP	Destatis 61111 / 42153	National CPI and industry production indices; the same calendar transformations and reference lags.
UK and German stress	ECB CLIFS	Country equity, bond and FX stress composites at reference lag two; not equivalent to a Treasury yield, VIX or Baa spread.
Five country themes	JKP Global Factor Data	Monthly ret in decimal USD; value-weighted characteristic portfolios averaged within signed themes. Theme memberships vary; these are conceptual counterparts to US factors.
UK three-month bills	Bank of England IUMAAJNB	Series discontinued in July 2017. An earlier rate file establishes the coverage constraint; it is excluded from this three-predictor experiment and not redistributed.
Other planned candidates	UK unemployment / AQR	Listed in the original source plan but no retained raw file or executable analysis. Their listing documents a plan, not completed data collection or evidence.

Eight international raw files supply the four retained input groups; exact file names, columns, monthly coverage, hashes and primary URLs are in documentation/data_source_decisions.csv. Reference lags are transparent timing proxies, not verified historical release dates. Original download timestamps and decision-month vintages are unavailable. The bill-series constraint does not establish that CLIFS was chosen as its replacement. Data descriptions are supported by ONS, Destatis, ECB, JKP and Bank of England sources cited in the report.

Documented candidates are distinguished from included inputs. The archived UK Treasury-bill candidate ends in June 2017, consistent with the Bank of England's discontinuation notice (Bank of England, n.d.); it does not supply a continuous short-rate predictor through evaluation. UK unemployment and AQR portfolios were planned alternatives, with no retrieved raw files retained in the source record. No results from those planned alternatives enter the reported comparisons.

Table B3 International Sharpe differences by block length

Paired comparison	Δ SR	3 months	6 months	12 months
UK Macro − mean	0.0602	[−0.0598, 0.2714]	[−0.0699, 0.2644]	[−0.0373, 0.2751]
UK State − Macro	−0.0973	[−0.4704, −0.0013]	[−0.5283, −0.0020]	[−0.5561, 0.0005]
DE Macro − mean	−0.00029	[−0.00403, 0.00246]	[−0.00404, 0.00255]	[−0.00381, 0.00266]
DE State − Macro	−0.0186	[−0.0560, 0.0100]	[−0.0569, 0.0107]	[−0.0537, 0.0100]

95% marginal fixed-path moving-block percentile-t intervals, 3,000 paired draws per setting, HAC3 studentizer. UK N=133; Germany N=122. Six-month blocks are primary. UK State−Macro intervals exclude zero at blocks three and six but include it at twelve; Sharpe intervals have no multiplicity or prior-selection correction. International two-sided mean tests use a separate four-comparison Holm family at each HAC lag and are not equivalent to these Sharpe intervals. Evaluation state counts, in Slowdown / Stagflation / Expansion / Overheating order, are UK 38 / 53 / 30 / 12 and Germany 31 / 61 / 18 / 12. The inherited international model omits Slowdown and selects separate Macro and state penalties. Full parameter, risk-date and state audits are in the mapped CSVs.

All transferred allocations use available JKP returns from the 36 natural months ending at the signal, the same positive-score normalization, shrinkage and cap, and a free initial entry followed by the 10 bp one-way target-weight charge. Forecasts and weights remain fixed in 3,000 paired bootstrap draws. Six-month blocks are primary, with three and twelve as sensitivities and a three-lag HAC studentizer. Mean-return Holm adjustments cover four international country–contrast comparisons separately at each HAC lag, without correcting the wider retrospective search. Zero-count states have undefined conditional means and do not become evidence of a zero effect. Evaluation-state counts are UK 38/53/30/12 and Germany 31/61/18/12 for Slowdown/Stagflation/Expansion/Overheating. All four states occur, although twelve Overheating months in each country provide limited support.

Appendix C IP input and penalty sensitivities

The common-penalty comparison uses the historical-IP calendar with the current-IP initial-training Ridge and EN parameters. Scalers and coefficients are refitted on exactly the historical-IP eligible labels. The sequential difference separates an input substitution under a common fitting policy from subsequent retuning; its allocation mapping, factor returns, fallback rules and costs stay fixed. This is computational attribution in the stated order, not a causal estimate.

Table C1 IP input and penalty comparisons

N	Input and estimation setting	Mean gain pp	Δ Sharpe	95% interval
175	Current IP / common penalties	0.300	0.055	[−0.083, 0.145]
175	Historical IP / common penalties	0.445	0.088	[−0.016, 0.177]
175	Historical IP / retuned penalties	0.449	0.089	[−0.015, 0.178]
88	Current IP / common penalties	0.364	0.041	[−0.155, 0.158]
88	Historical IP / common penalties	0.656	0.094	[−0.042, 0.209]
88	Historical IP / retuned penalties	0.664	0.096	[−0.041, 0.210]

Macro minus its identically mapped historical mean, primary five-factor allocation, 10 bp sleeve charges. Common penalties are the current-IP initial-training choices, applied unchanged to both input calendars. The historical-IP retuned design uses its own initial-training choices. Sharpe intervals use 3,000 paired six-month moving-block draws and HAC3 studentization. Each contrast conditions on its generated paths.

Table C2 Mean-return dependence sensitivities over 175 months

IP setting	HAC lag	Raw p	Portfolio-family Holm p	IP-family Holm p
Current IP	3	0.2923	0.8769	0.2923

IP setting	HAC lag	Raw p	Portfolio-family Holm p	IP-family Holm p
Current IP	6	0.2486	0.7457	0.2486
Current IP	12	0.1954	0.5862	0.1954
Historical IP / common	3	0.0678	—	0.1981
Historical IP / common	6	0.0414	—	0.1210
Historical IP / common	12	0.0163	—	0.0473
Historical IP / retuned	3	0.0660	0.1981	0.1981
Historical IP / retuned	6	0.0403	0.1210	0.1210
Historical IP / retuned	12	0.0158	0.0473	0.0473

Two-sided tests of zero expected monthly Macro-minus-mean return. Portfolio-family Holm adjusts three contrasts within each existing design and lag: Macro minus mean, fixed state minus Macro, expanding state minus Macro. IP-family Holm instead adjusts the three Macro-minus-mean input/penalty settings within each cohort and lag. The common-penalty IP supplement has no state fits, so its portfolio-family entry is undefined. Neither correction covers selection across HAC lags or the complete specification search. HAC3 is primary; HAC6 and HAC12 are sensitivities. Corresponding 88-month values are in the mapped result file.

Appendix D Recorded specification and event diagnostics

Each of the ten documented mappings compares Macro with an identically mapped historical-mean allocation over 175 common targets at a 10 bp sleeve charge. They are reconstructed settings rather than a complete experiment history. The Reality Check conditions on fitted paths; its secondary family adds both state models under each mapping. Event exclusions retain original weights and charges.

Table D1 All documented allocation mappings

Mapping	Mean SR	Macro SR	Δ Sharpe	Mean gain pp	Turnover %
Six factors / raw	0.958	0.951	−0.008	1.095	13.8
Six factors / raw / 5% ceiling	0.948	0.921	−0.026	−0.081	14.3
Six factors / variance-scaled	0.678	0.703	0.025	0.258	5.9
Six factors / variance / 5% ceiling	0.685	0.710	0.025	0.210	6.4
Six factors / ranks	0.902	0.812	−0.090	−0.379	5.4
Six factors / ranks / 5% ceiling	0.924	0.820	−0.105	−0.446	6.6
Five factors / raw	0.173	0.393	0.221	1.542	13.8
Five factors / variance (primary)	0.203	0.258	0.055	0.300	5.5
Fixed market 1/6 / raw	0.720	0.830	0.109	1.285	11.5
Fixed market 1/6 / variance	0.757	0.790	0.033	0.250	4.6

175 targets, current-vintage forecasts, net 10 bp allocation charges. Variance and rank mappings use 50% equal-weight shrinkage and a 40% cap; raw mappings normalize positive forecasts. A 5% annual-volatility ceiling scales risky weights downward, with residual zero-excess-return cash. Fixed-market mappings reserve 1/6 for MKT−RF. Mean gain is Macro minus the identically mapped historical mean in annual pp. Reality Check p-values for 3 / 6 / 12-month blocks: Macro ten-mapping family 0.101 / 0.090 / 0.095; thirty Macro/state combinations 0.104 / 0.123 / 0.140. Paired non-circular moving blocks, 9999 draws, unstudentized maximum of centered mean gains. The primary mapping's maximum-adjusted p is 0.520 at six-month blocks. The correction covers these documented reconstructed mappings, conditional on fitted paths. Turnover is Macro's mean monthly one-way target-weight change, in %, with free initial entry. Mappings can change scaling and constraints jointly.

Table D2 Initial continuous-model complexity

Factor	Ridge penalty	Ridge slope df	EN penalty	EN active slopes
MKT−RF	10^6	0.0018	1.5849	0
SMB	562.34	1.9318	2.5119	2
HML	10^6	0.0018	1.5849	0
RMW	10^6	0.0018	0.6310	0
CMA	10^6	0.0018	1.0000	0
MOM	316.23	2.6844	1.5849	0

261 initial current-vintage training signals. Ridge slope df is trace$[Z(Z'Z + \alpha I)^{-1}Z']$, for the centered standardized design and fixed penalty; add one for the unpenalized intercept. EN active coefficients use a 10^{-10} threshold and are a complexity proxy, not an exact effective-df estimate. EN mixing ratios and all refit coefficients remain in the numerical outputs.

Table D3 Same COVID episode under two allocation rules

Mapping	N	Macro sum pp	Mean sum pp	Difference pp
Five factors / variance (primary)	10	−4.47	−3.11	−1.36
Six factors / raw	10	25.19	9.22	15.97

March–December 2020, current-vintage inputs, 10 bp allocation charges. These are sums of monthly arithmetic returns and paired differences. They are distinct from separately compounded portfolio returns or a difference in compounded wealth.

Appendix E Initial associations and RMW state averages

Table E1 Initial-training OLS slope associations

Predictor	MKT−RF	SMB	HML	RMW	CMA	MOM
VIX	0.111	−0.097*	−0.122*	−0.046	−0.058	0.007
Credit spread	−0.821	1.433†	1.042	0.327	0.656	−2.291
Term spread	−0.450	0.383	0.011	−0.053	−0.058	1.408
2Y yield change	0.678	0.684	0.324	−1.369	0.486	0.762
CPI inflation	1.214	0.190	2.260*	0.738	0.184	−2.532
Unemployment	0.586	−0.295	−0.175	−0.031	−0.090	−0.682
IP growth	0.782	0.294	0.115	−0.297	−0.033	−0.597

261 initial training signals, next-month factor return in percentage points. Coefficients use raw predictor units: one VIX point, one unemployment percentage point, or one percentage point of the specified growth, inflation, spread or yield change. * denotes two-sided HAC3 $p<0.05$ before correction; † denotes Holm $p<0.05$ across all 42 slopes. Four raw slopes and one corrected slope pass 5%; the surviving association is SMB with credit spread. These are initial-sample conditional associations, not causal channels or out-of-sample skill. Cross-validation selects penalties using forecast loss, not these p-values.

Table E2 Observed next-month RMW returns by state

Rule	State	Training N / mean %	Former 87 N / mean %	Original 88 N / mean %	Extended 175 N / mean %
Fixed	Expansion	75 / −0.357	21 / −0.309	0 / —	21 / −0.309
Fixed	Overheating	55 / 0.924	5 / −0.218	11 / 2.041	16 / 1.335
Fixed	Slowdown	56 / 0.434	59 / 0.086	39 / −0.087	98 / 0.017
Fixed	Stagflation	75 / 0.921	2 / 0.720	36 / 0.116	38 / 0.148
Expanding	Expansion	75 / −0.357	22 / −0.387	0 / —	22 / −0.387
Expanding	Overheating	55 / 0.924	7 / −0.070	11 / 2.041	18 / 1.220
Expanding	Slowdown	56 / 0.434	56 / 0.082	31 / −0.036	87 / 0.040
Expanding	Stagflation	75 / 0.921	2 / 1.660	44 / 0.044	46 / 0.114

Cells give the number of state-labeled targets and the mean subsequent monthly operating-profitability factor return in %. Training targets end April 1990–December 2011; former validation January 2012–March 2019; original evaluation April 2019–July 2026. Two state labels are missing in each later evaluation window; state-labeled months may include a continuous-model fallback, so the original 86 labeled targets differ from 85 matched forecast origins. An absent state has N=0 and an undefined mean (—). Means are descriptive conditional associations; small unequal cells do not identify a defensive or causal mechanism. Dispersion and continuous-model eligibility counts remain in the mapped CSV.

Appendix F AI-use disclosure

Artificial intelligence (AI) tools were used to assist with code development and debugging in the preparation of this report. I was responsible for the literature search, retrieval, and review; study design; data collection and analysis; interpretation of the results and formulation of conclusions; and preparation of the final report.